

SLI 97CS(I)(N)FXxxxxPE PG-VQFN-32-13

Component Qualification Report for Packaged Integrated Circuits

based on AEC Q100 Rev.H

About this document

Scope and purpose

This document provides information about the results of the product(s) qualification.

Intended audience

This Qualification Report is intended for customers of Infineon Technologies and certification parties. The data contained in this document is exclusively intended for technically trained staff.



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General Information

1 General Information

Table 1 General Information

Product Description	SOLID FLASH™ Controller for Security Applications
Device Family	In-car Security Controller (SLI)
Wafer Process Technology	L90FL
Wafer Diameter	200 mm
Wafer Production Site(s)	Infineon Technologies (IFX), Dresden, Germany
Pre-Assembly Site(s)	Same as Assembly Site(s)
Assembly Site(s)	Amkor Technology, Laguna, Philippines
Part Operating Temperature	Grade 2: -40°C to 105°C
MSL	MSL1
Applied Qualification Standards	AEC Q100 Rev. H
PCN / Info Note	036/19

2 Family Qualification with Generic Data from Reference Parts / Design / Products

Table 2 Family Qualification

	Product (Identifier)	Design step	Wafer Technology	Wafer Production Site	Package	Assembly Site
Part to be Qualified	SLI 97CS(I)(N)FXxxxPE	A11	L90FL	IFX	PG-VQFN-32-13	Amkor
Reference part A	SLI 97C(U)(N)(F)(X)xxxPE	A11	L90FL	IFX	PG-VQFN-8-4	Amkor
Reference part B	SLI 9670AQ2.0	A11	L90FL	IFX	PG-VQFN-32-13	Amkor

3 Qualification Result Overview

Table 3 Qualification Result Overview

TEST GROUP A – ACCELERATED ENVIRONMENT STRESS TESTS							
TEST #	STRESS TEST according AEC Q100 Rev. H	Test Conditions	# LOTS	Sample Size	Part to be Qualified	Ref part A	Ref Part B (1 lot)
-	Moisture Sensitivity Level	MSL: 1 peak temp.: 260°C J-STD-020		64	PASS		
A1	Preconditioning	MSL: 1 peak temp.: 260°C J-STD-020, JESD22-A113	3	231		PASS	PASS
A2	Temperature Humidity Bias	85°C/85%RH/1000h JESD22-A101	3	77	PASS *	PASS	PASS
A3	uHAST	130°C / 85% r.h. 230kPa 96h, JESD22-A118	3	77	PASS *	PASS	PASS
A4	Temperature Cycling	-55/125°C, 1000x JESD22-A104	3	77	PASS *	PASS	PASS
A5	Power Temperature Cycling	n.a.			not applicable		
A6	High Temperature Storage Life	150°C, 500h JESD22-A103	1	45	PASS		
TEST GROUP B – ACCELERATED LIFETIME SIMULATION TESTS							
B1	High Temperature Operating Life	Tj=125°C, 1000h, JESD22-A108	3	77		PASS	
B2	Early Life Failure Rate	Tj=125°C AEC Q100-008	3	800		PASS	
B3	NVM Endurance, Data Retention, & Operational Life	P/E cycles: 100k and 500k HTSL: 150°C/500h HTOL: 125°C/408h AEC Q100-005	3	77		PASS	
							Reference part A has been qualified acc AEC Q100 Rev G

No delamination after Scanning Acoustic Microscope (SAM)

* Part to be qualified was qualified as MSL3.
After delta qualification it is re-classified as MSL1

SLI 97 is no power chip, rated < 1W

TEST #	STRESS TEST according AEC Q100 Rev. H	Test Conditions	# LOTS	Sample Size	Part to be Qualified	Ref part A	Ref Part B (1 lot)	Remarks
TEST GROUP C – PACKAGE ASSEMBLY INTEGRITY TESTS								
C1	Wire Bond Shear	AEC Q100-001	1	5	PASS			after TC
C2	Wire Bond Pull	MIL-STD-883 - 2011	1	5	PASS			after TC
C3	Solderability	JESD22-B102	1	15	PASS			
C4	Package Dimensions	JESD22-B100/108 acc. package specifications	3	10	PASS			
C5	Solder Ball Shear				not applicable			no solder balls on VQFN-32
C6	Lead Integrity				not applicable			no solder balls on VQFN-32
TEST GROUP D – DIE FABRICATION RELIABILITY TESTS								
D1	Electromigration				PASS			part of technology qualification
D2	Time Dependent Dielectric Breakdown				PASS			part of technology qualification
D3	Hot Carrier Injection				PASS			part of technology qualification
D4	Negative Bias Temperature Instability				PASS			part of technology qualification
D5	Stress Migration				PASS			part of technology qualification
TEST GROUP E – ELECTRICAL VERIFICATION TESTS								
E1	Pre- and Post-Stress Function Parameter	Test @ CT, RT, HT acc to AEC Q100	all	all	PASS	PASS	PASS	
E2	Electrostatic Discharge Human Body Model	2kV (all interfaces) AECQ100-002	1	3/voltage		PASS		
E3	Electrostatic Discharge Charged Device Model	750V AEC Q100-011	1	3/voltage	PASS			
E4	Latch-Up	±150mA, 105°C AECQ100-004	1	6		PASS		

Qualification Result Overview

E5	Electrical Distributions	AEC Q100-009	3	30	PASS			on wafer level and IFX package (CDIP)
E6	Fault Grading	AEC Q100-007						Only static stuck-at fault model applied (IDDQ not available by design)
E7	Characterization	AEC Q003	3	30	PASS			on wafer level and IFX package (CDIP)
E9	Electromagnetic Compatibility	based on SAE J1752/3	1	1		PASS		IFX internal procedure, aligned for GSM operation
E10	Short Circuit Characterization				not applicable			SLI 97 is no smart-power device
E11	Soft Error Rate				not applicable			SLI 97 contains SRAM < 1MByte

TEST GROUP F – DEFECT SCREENING TESTS

F1	Process Average Testing							Implemented for mode currents (IDDQ not available by design)
F2	Statistical Bin/Yield Analysis							implemented for all softbins

TEST GROUP G – CAVITY PACKAGE INTEGRITY TESTS

G1	Mechanical Shock	JESD22-B104 5 pulses, 0.5 ms, 1500g	1	15			PASS	No cavity package As information for ETSI compliance
G2	Variable Frequency Vibration	JESD22-B103 20-2000Hz	1	15			PASS	No cavity package As information for ETSI compliance
G3	Constant Acceleration				not applicable			No cavity package
G4	Gross/Fine Leak				not applicable			No cavity package
G5	Package Drop				not applicable			No cavity package
G6	Lid Torque				not applicable			No cavity package
G7	Die Shear Strength				not applicable			No cavity package
G8	Internal Water Vapor				not applicable			No cavity package

Revision History

Revision History

Reference	Description
Revision 1.2, 2020-07-03	
All	General review
Revision 1.1, 2019-07-29	
All	New template. General review. Change from MSL3 to MSL1.
Revision 1.0, 2015-10-21	
All	Initial version

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Edition 2020-07-03

Published by

Infineon Technologies AG
81726 München, Germany

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